

AEGIS

The Physics of Optical Hidden-Camera Detection: Lens–Sensor Retro-Reflection

The Core Idea

Our optical subsystem doesn't rely on ordinary reflection off the outside of a lens — it exploits lens–sensor retro-reflection, also known as the cat-eye effect. IR light that enters a hidden camera's lens reaches the image sensor, reflects, and travels back out through the same lens toward the source, rather than scattering away like a normal surface.

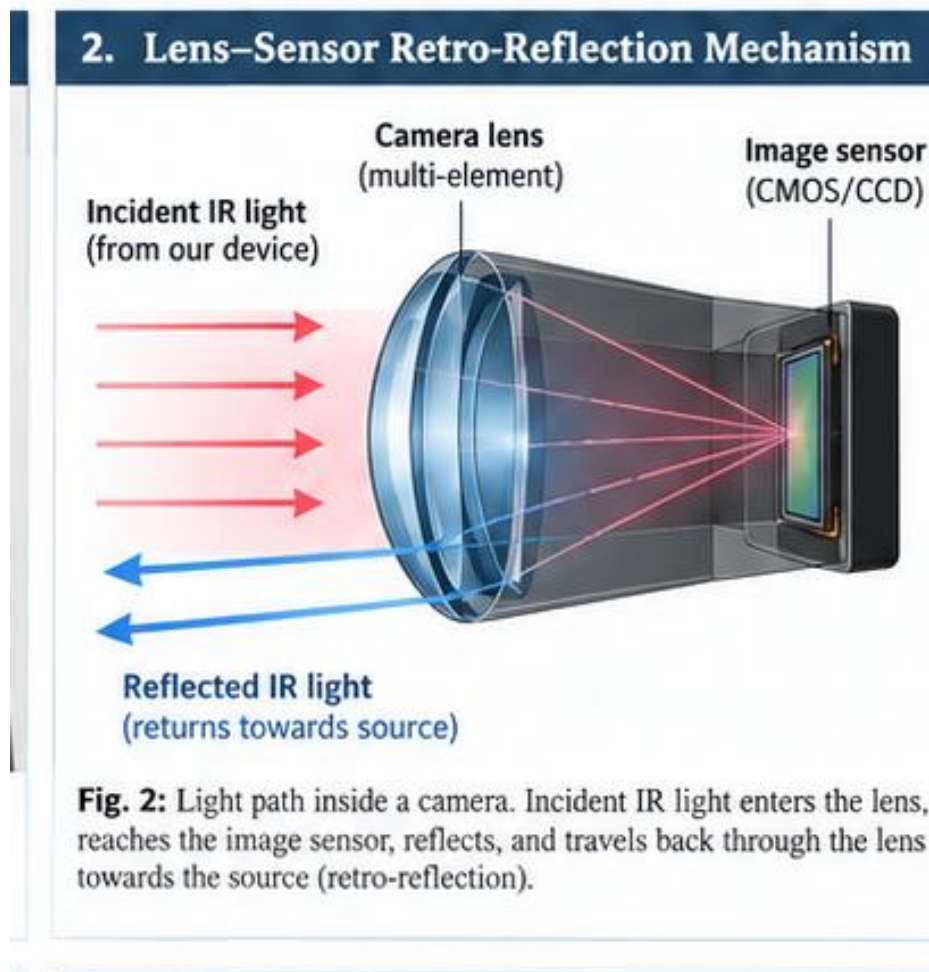


Fig. 1 — IR light enters the lens, reflects off the image sensor, and exits back toward the source.

Why This Differs From Simple Reflection

A convex lens focuses incoming rays onto the sensor; the sensor reflects a fraction of that light; the same lens then redirects it back out roughly parallel to the incident beam. This is why a co-located IR source and detector — not a source and detector at different angles — can pick up the return signal.

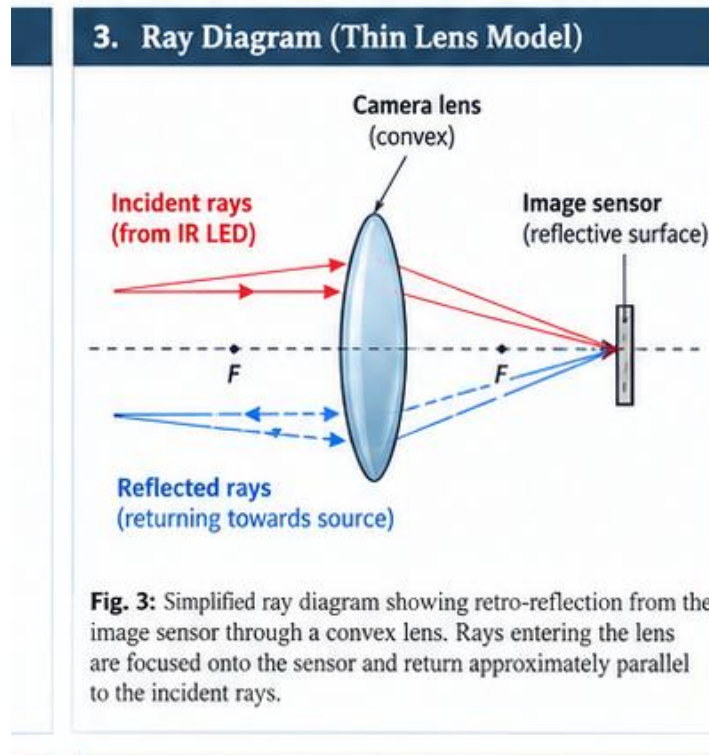


Fig. 2 — Thin-lens ray diagram: incident and reflected rays stay nearly parallel.

Detection Geometry

Because the IR LED and phone camera sit right next to each other, the device can capture its own retro-reflected light. The effect is only visible within a narrow angular window ($\approx 20^\circ$, observed experimentally) and its position relative to the hidden camera matters.

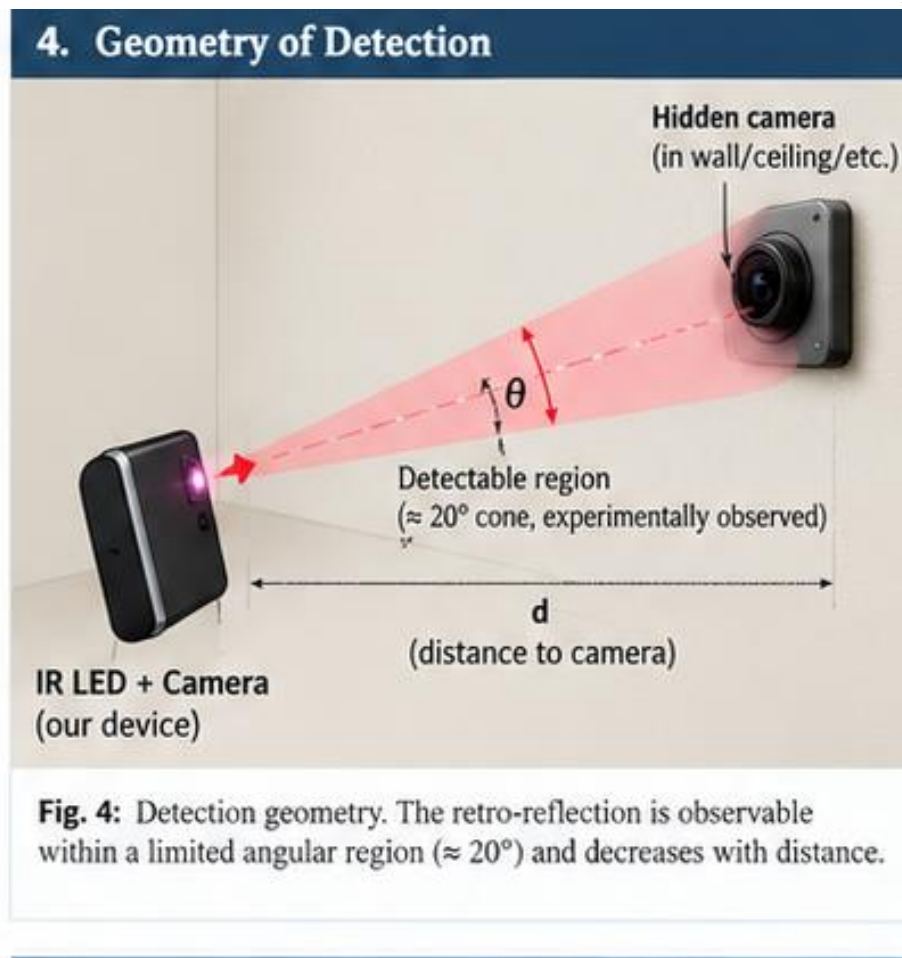


Fig. 3 — Detection cone between the AEGIS device and a hidden camera.

The Governing Relationships

$$I \propto 1 / d^2$$

Returned intensity I falls off with the square of the distance d to the hidden camera, and the retro-reflection is only detectable inside a roughly 20° field of view around the direct line of sight — both of which set practical limits on scanning range and angle.

7. Intensity vs. Distance (Inverse-Square Law)

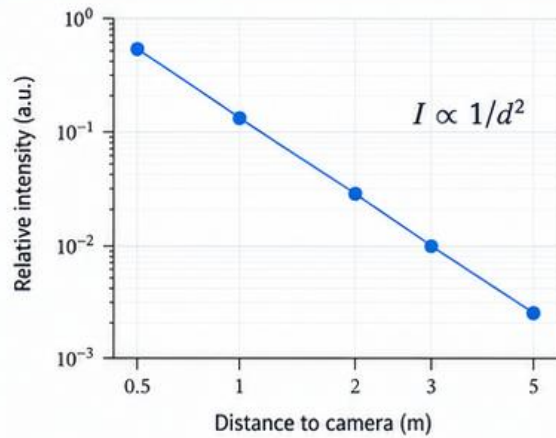


Fig. 7: Received reflection intensity decreases approximately with the inverse square of the distance ($I \propto 1/d^2$).

Fig. 4 — Intensity vs. distance (inverse-square law).

8. Field of View for Detection

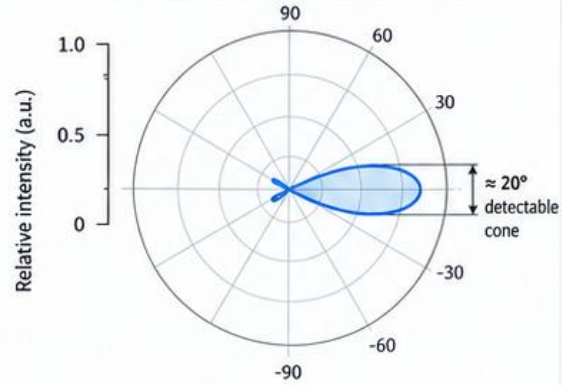


Fig. 8: Measured angular response (representative). The retro-reflection is typically visible within a limited cone ($\sim 20^\circ$).

Fig. 5 — Angular response: detectable cone $\approx 20^\circ$.

Detection Pipeline

In practice: IR illumination hits the hidden camera, the lens–sensor system retro-reflects it, the phone camera captures the return as a bright localized spot, and image processing flags it as a suspected camera.

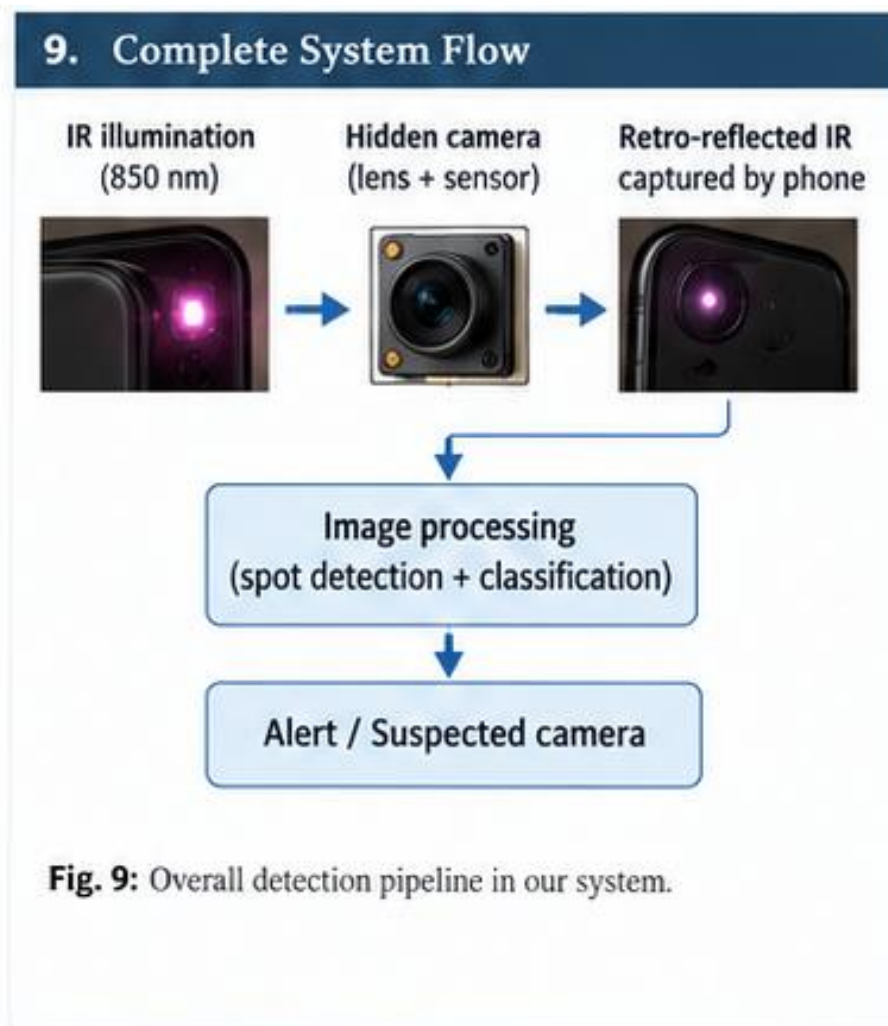


Fig. 6 — End-to-end optical detection flow.

Important Limitation

A bright IR reflection alone doesn't confirm a hidden camera — mirrors, glass, metal, and other glossy surfaces reflect too. Retro-reflection is a candidate-generation signal; distinguishing true camera-lens signatures from ordinary reflective surfaces is the open problem, and the reason we plan to add computer-vision classification on top of the raw IR spot detection.

Reference

S. Sami, S. R. X. Tan, B. Sun, and J. Han, "LAPD: Hidden Spy Camera Detection using Smartphone Time-of-Flight Sensors," *Proceedings of the 19th ACM Conference on Embedded Networked Sensor Systems (SenSys 2021)*, pp. 288–301, 2021. DOI: 10.1145/3485730.3485941.

This paper establishes the lens–sensor retro-reflection phenomenon experimentally; AEGIS applies the same underlying physics with a dedicated IR LED and phone camera rather than a ToF sensor.